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Candidate Number

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HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION BYTOPIC

MATHEMATICS Extended Part

Module 2 (Algebra and Calculus)

Question-Answer Book

Time allowed : $2\frac{1}{2}$ hours

This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.
- (2) This paper consists of TWO sections, A and B.
- (3) This paper carried 100 marks.
- (4) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (6) Unless otherwise specified, all working must be clearly shown.
- (7) Unless otherwise specified, numerical answers must be exact.
- (8) The diagrams in this paper are not necessarily drawn to scale.
- (9) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

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FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$ $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$ $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$	$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$ $\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$ $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$ $\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$
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SECTION A (56 marks)

1. (a) Prove that $1 + \frac{2 \sin 2x - \cos 3x}{\cos x} = 4 \sin x (1 + \sin x)$ for all x where $\cos x \neq 0$. (3 marks)
- (b) Solve $2 + \frac{2 \sin 2\theta - \cos 3\theta}{\cos \theta} = 0$, where $-\pi \leq \theta \leq \pi$. (3 marks)

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3. (a) Prove that $\tan x + 2 \tan 2x = \cot x - 4 \cot 4x$. (3 marks)

(b) Solve $\tan x + 2 \tan 2x + 4 \tan 4x + 8 \tan 8x - \cot x = 16$, where $\frac{\pi}{10} < x < \frac{\pi}{8}$. (3 marks)

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4. It is given that $0 \leq \theta \leq \frac{\pi}{2}$ and $\sin 2\theta = \frac{2}{3}$.

(a) Without finding the value of θ , prove that

$$\tan^2 \theta + \cot^2 \theta = 7.$$

(3 marks)

(b) Find the exact value of

$$\frac{1}{1 + \tan^2 \theta} + \frac{1}{1 + \cot^2 \theta} - \frac{12}{\tan^4 \theta + \cot^4 \theta - 2}.$$

(3 marks)

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5. (a) Expand $(a + b + c)^2$. (1 mark)

(b) Let α, β, γ be real numbers such that $\cos \alpha + \cos \beta + \cos \gamma = 0$ and $\sin \alpha + \sin \beta + \sin \gamma = 0$.

(i) Find the value of $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha)$. (2 marks)

(ii) Prove that $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = 0$. (3 marks)

(c) Let ϕ be a real constant. Find the value of

$$\sin(\alpha - \phi) \cos(\alpha - \phi) + \sin(\beta - \phi) \cos(\beta - \phi) + \sin(\gamma - \phi) \cos(\gamma - \phi).$$

(3 marks)

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$$\tan^6 \theta - 33 \tan^4 \theta + 27 \tan^2 \theta - 3 = 0.$$

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7. Let n be a positive integer. It is given that for all real numbers θ satisfying $\sin \theta \neq 0$,

$$\frac{\sin(2n+1)\theta}{\sin \theta} = 1 + 2 \sum_{k=1}^n \cos(2k\theta).$$

(a) Prove that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$. (2 marks)

(b) (i) Prove that $\cos 3x = 4 \cos^3 x - 3 \cos x$. (2 marks)

(ii) By considering the equation $\cos 3\theta = \cos 4\theta$, prove that $8 \cos^3 \theta + 4 \cos^2 \theta - 4 \cos \theta - 1 = 0$ holds for $\theta = \frac{2\pi}{7}$. (3 marks)

(iii) Hence, without using a calculator, find the value of $\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$. (2 marks)

(c) Using the results of (a) and (b), find the value of each of the following without using a calculator:

(i) $\cos^2 \frac{2\pi}{7} + \cos^2 \frac{4\pi}{7} + \cos^2 \frac{6\pi}{7}$ (2 marks)

(ii) $\sec \frac{2\pi}{7} + \sec \frac{4\pi}{7} + \sec \frac{6\pi}{7}$ (2 marks)

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SECTION B (44 marks)

8. (a) By mathematical induction, prove that for all positive integers n ,

$$\sum_{k=1}^n \frac{1}{2^k} \tan \left(\frac{\theta}{2^k} \right) = \frac{1}{2^n} \cot \left(\frac{\theta}{2^n} \right) - \cot \theta,$$

where $\theta \neq m\pi$ (m is an integer). (4 marks)

- (b) Hence, find the exact value of

$$\frac{1}{32} \tan \left(\frac{\pi}{96} \right) + \cdots + \frac{1}{1024} \tan \left(\frac{\pi}{3072} \right).$$

(2 marks)

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9. (a) Prove that $\tan\left(\frac{\pi}{4} - \theta\right) = \frac{\cos 2\theta}{1 + \sin 2\theta}$, where $\sin 2\theta \neq -1$ and $\cos \theta \neq 0$. (2 marks)

(b) Let ϕ be a real number. Prove that for all positive integers n ,

$$\sum_{r=1}^n C_r^{n+1} (-1)^r (\sin 2\phi)^r (1 + \sin 2\phi)^{n+1-r} = 1 - (1 + \sin 2\phi)^{n+1} - (-\sin 2\phi)^{n+1}.$$

(2 marks)

(c) Let $\phi = \frac{\pi}{8}$. Using the results of (a) and (b), find the exact value of the series

$$\sum_{r=1}^3 C_r^4 \left(-\frac{1}{2}\right)^r (\sin 4\phi)^r (\cos 2\phi)^{4-2r} \left[\tan\left(\frac{\pi}{4} - \phi\right)\right]^{r-4}.$$

(3 marks)

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10. It is given that θ is an acute angle. Let $I = \int \frac{1}{\sin \theta(2+\cos \theta)} d\theta$.

(a) Prove that

$$\frac{1}{\sin \theta(2+\cos \theta)} = \frac{\sin \theta}{6(1-\cos \theta)} + \frac{\sin \theta}{2(1+\cos \theta)} - \frac{\sin \theta}{3(2+\cos \theta)}.$$

(2 marks)

(b) Prove that

$$I = \frac{1}{3} \ln \left(3 \tan \frac{\theta}{2} + \tan^3 \frac{\theta}{2} \right) + C,$$

where C is an arbitrary constant.

(4 marks)

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11. Let $f(x) = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, where $0 < x < \frac{\pi}{4}$.

(a) Prove that $f(x) = \frac{\cos x - \sin x}{\cos x + \sin x}$. (2 marks)

(b) From first principles, find $f'(x)$. (4 marks)

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13. Let $M = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, where $0 < \theta < \frac{\pi}{2}$.

It is given that $\det(M^2 - \sqrt{3}M + I) = 0$.

- (a) Prove that $2 \cos 2\theta - 4\sqrt{3} \cos \theta + 5 = 0$. (3 marks)
- (b) Hence, find the exact value of θ . (3 marks)
- (c) Let M be the matrix with the value of θ found in (b). A sequence of matrices is defined by $M^n = a_n M + b_n I$ for all positive integers n , where a_n and b_n are real constants. Find the exact value of a_{2026} . (2 marks)

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